UNIT-5(Linux and Unix Case Study)
Linux Introduction

LECTURE NO:-1

Linux Introduction

Linux is the world's most dominant operating system.
Launched in 1991 by Linus Torvalds, it's the gold standard of user-led open source innovation, representing Linus' desire for an OS that he could run on his personal computer.

Since those early days, developers from large hardware companies to small emerging technology providers have contributed to Linux, building solutions that run on top of the open-source OS.

Between 2005 and 2016, more than 14,000 individual developers from more than 1,300 companies contributed to the Linux kernel project. In an industry that features fierce competition between companies, it's amazing to see a project with common, shared resources developed on a massive scale.

Regular two-to-three month releases deliver stable updates to Linux users, adding significant new features, enhanced device support, and improved performance. The rate of change in the kernel has been historically high, and continues to increase – more than 10,000 patches often go into each kernel release. Each release contains the work of more than 1,600 developers, representing 200+ companies worldwide.
But this cooperation isn't just limited to the Linux kernel. In 2005, the world saw the rise of the powerful Git revision control tool.

When BitKeeper no longer met the Linux kernel community's need for a distributed system, Linus Torvalds took the challenge into his own hands. Disappearing over the weekend to build a new tool, he emerged with Git, software that has ushered in a new level of social coding among developers. Git forms the basis of GitHub, the single largest open-source repository online today.
Together, Linux and Git – both created by Linus Torvalds – are the two most highly rated open-source projects in history. But let's step back to the early days of this projec

The Challenge -

After it was created in AT&T's Bell Laboratories in the 1970s, the UNIX operating system saw success as a powerful, and initially free, operating system. Through the 1980s, the market for UNIX systems had grown steadily, but had become fragmented. Arguments arose as vendors slowly tied the OS and features to proprietary systems.
In the 1990s, the open source community was becoming more popular, and the costs dropped for x86 processors on desktops and servers. Internet usage was rising, and the appetite for web consumption became insatiable. Companies were tasked with providing email and web content on servers that could handle exceptional growth. Online commerce (e.g., Black Friday) and global events (e.g., World Cup soccer) placed extraordinary demands upon web servers.
The Solaris operating system, which sat on the Sun SPARC system based on the PowerPC platform, was lauded for its scalability and security. At the same time, an industry weary from fighting about the discrepancies of different UNIX systems saw promise elsewhere – in the newly created Linux OS. Combined with the gaining momentum of the open source movement, an industry saw a new opportunity.
By the mid-1990s, Linux started to gain traction, most notably in web hosting, network and simple database applications, hosted on open source MySQL and Postgres. However, the use of Linux for finance and other mission-critical systems was modest at best. As the importance of the web grew, so did the need for a strong level of standards, free of dominance by any one vendor. The Apache Web Server was born, providing a level playing field for the web, driving a de facto web standard. The extensibility of Apache via compiled modules let many developers create new features concurrently without requiring a highly coordinated or centralized development process.

The Approach

As personal web usage increased, Linux continued its steady development. People hosting web content needed an OS that could handle exponential traffic growth without seeing their expenses rise exponentially. The cost of proprietary systems was high, making them increasingly unattractive.
Linux offered utility in a world where server operating systems were at a premium. Price, performance and flexibility on how to deploy and modify was all better than proprietary systems. There were three keys to the Linux success story:

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"It didn't even start by people contributing code, it was more that people started contributing ideas. And just the fact that somebody else takes an interest in your project – and I'm sure it's true of other things, too, but it's definitely true in code—and looks at it

enough to actually give you feedback and give you ideas. That was a huge thing for me."

– Linus Torvalds, Linux creator

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UNIT-5(Linux and Unix Case Study)
Result
LECTURE NO:-2

Result:

From its beginning, when a young Finnish grad student revolutionized the computing world, Linux has become the ultimate Cinderella story.
Everyone loves an underdog, but that chapter has closed. Linux is now the OS of choice among users worldwide – open source is the default development methodology for users bringing solutions to market. Linux has spread its wings to industries and markets around the globe, including.

CARRIER DATA CENTERS

The Carrier Grade Linux (CGL) initiative launched in 2002, with UNIX dominating the data center marketplace. Solaris hosted more than 95% of telecommunications applications.
To qualify as a "carrier grade" platform, Linux needed to cross gaps in performance, API standards, scalability, reliability, availability and tooling. Undaunted, the CGL workgroups tackled these challenges incrementally, enlisting players from across the IT industry, who would also benefit from new capabilities. Within five years, Linux was competing on an equal footing with legacy UNIX for telecoms. Today, Linux is considered the best option for carrier data center and cloud deployment.

FINANCIAL MARKETS

Speed matters in the world of stock trading and financial transactions. Once tied to proprietary systems with long release cycles, today's financial exchanges — NASDAQ, the London Exchange, the Tokyo Stock Exchange and a majority of others — now run on Linux.
The New York Stock Exchange Euronext, the largest exchange, runs on a Linux system that can generate 1.5 million quotes and process 250,000 orders every second, offering acknowledgements of each transactions within two milliseconds.
In addition to speed, the ability to customize and a revamped scheduler that keeps processes such as trades from being interrupted, have made Linux the gold standard in the financial industry.

CLOUD COMPUTING

More than half of senior enterprise IT executives expect the majority of IT workloads to reside in the cloud or co-location site in the future, according to a recent survey by the Uptime Institute. Of those, 70% expect the shift to happen in the next four years.
With more than 75% of cloud-enabled enterprises running Linux as their primary cloud platform, we expect to see a dramatic increase in Linux deployments to support the move to the cloud.

SMARTPHONES

In 2014, the Linux-based Android smartphone OS was shipped on more than 1 billion units, eclipsing the previous year's total smartphone sales.

SUPERCOMPUTING

Once dominated by highly specialized hardware and operating systems, Linux has become dominant in the supercomputing market. According to a November 2016 report by Top 500 (which ranks systems based on industry standard benchmarks), 99.6% of the top 500 most powerful computing systems ran Linux, with commercial UNIX (specifically AIX) accounted for the other 0.4%.

EMBEDDED LINUX SUCCESS

Once a highly fragmented marketplace of more than 300 embedded operating systems, Linux now serves nearly every segment of intelligent device design. From industrial control to communications, networking, mobile/wireless, automotive, medical devices and beyond, Linux spans the gamut of embedded system types and form factors. Small, embedded systems, Internet of Things nodes, single-board computers and clusters of massively parallel blades are all benefiting from the Linux system.

From its humble beginning, Linux has grown into a vital technology, revolutionizing entire industries. We expect the open-source development model, proven successful by Linux, will continue to empower developers around the world to create similar communities to solve technical challenges together long into the future.

UNIT-4(File Management)
Unix Analysis

LECTURE NO:-3

UNIX Analysis:

Most operating systems can be grouped into two different families. Aside from Microsoft's Windows NT-based operating systems, nearly everything else traces its heritage back to Unix.

Linux, Mac OS X, Android, iOS, Chrome OS, Orbis OS used on the PlayStation 4, whatever firmware is running on your router — all of these operating systems are often called "Unix-like" operating systems.

Unix's Design Lives On Today

Unix was developed in AT&T's Bell Labs back in the mid-to-late 1960's. The initial release of Unix had some important design attributes that live on today.

One is the "Unix philosophy" of creating small, modular utilities that do one thing and do them well. If you're familiar with using a Linux terminal, this should be familiar to you — the system offers a number of utilities that can be combined in different ways through pipes and other features to perform more complex tasks. Even graphical programs are likely calling simpler utilities in the background to do the heavy lifting. This also makes it easy to create shell scripts, stringing together simple tools to do complicated things.

Unix also had a single file system that programs use to communicate with each other. This is why "everything is a file" on Linux – including hardware devices and special files that provide system information or other data. It's also why only Windows has drive letters, which it inherited from DOS — on other operating systems, every file on the system is part of a single directory hierarchy.

Tracing the Unix Descendants

Like any history going back over 40 years, the history of Unix and its descendants is messy. To simplify things, we can roughly group Unix's descendants into two groups.

One group of Unix descendants were developed in academia. The first was BSD (Berkeley Software Distribution), an open-source, Unix-like operating system. BSD lives on today through FreeBSD, NetBSD, and OpenBSD. NeXTStep was also based on the original BSD, Apple's Mac OS X was based on NeXTStep, and iOS was based on Mac OS X. Many other operating systems, including the Orbis OS used on the PlayStation 4, are derived from types of BSD operating systems.

Richard Stallman's GNU project was also started as a reaction to AT&T's increasingly restrictive Unix software licensing terms. MINIX was a Unix-like operating system created for educational purposes, and Linux was inspired by MINIX. The Linux we know today is really GNU/Linux, as it's made up of the Linux kernel and a lot of GNU utilities. GNU/Linux isn't directly descended from BSD, but it is descended from Unix's design and has its roots in academia. Many operating systems today, including Android, Chrome OS, Steam OS, and a huge amount of embedded operating systems for devices, are based on Linux.

On the other hand, there were the commercial Unix operating systems. AT&T UNIX, SCO UnixWare, Sun Microsystems Solaris, HP-UX, IBM AIX, SGI IRIX — many big corporations wanted to create and license their own versions of Unix. These aren't quite as common today, but some of them are still out there.

UNIT-4(File Management)
The Rise of DOS and Windows NT
LECTURE NO:-4

The Rise of DOS and Windows NT

Many people expected Unix to become the industry standard operating system, but DOS and "IBM PC compatible" computers eventually exploded in popularity. Microsoft's DOS became the most successful DOS of them all. DOS was never based on Unix at all, which is why Windows uses a backslash for file paths while everything else uses a forward slash. This decision was made back in the early days of DOS, and later versions of Windows inherited it, just as BSD, Linux, Mac OS X, and other Unix-like operating systems inherited many aspects of Unix's design.

Windows 3.1, Windows 95, Windows 98, and Windows ME were all based on DOS underneath. Microsoft was developing a more modern and stable operating system at the time, which they named Windows NT — for "Windows New Technology." Windows NT eventually made its way to regular computer users as Windows XP, but it was available for corporations as Windows 2000 and Windows NT before that.

All of Microsoft's operating systems are based on the Windows NT kernel today. Windows 7, Windows 8, Windows RT, Windows Phone 8, Windows Server, and the Xbox One's operating system all use the Windows NT kernel. Unlike most other operating systems, Windows NT wasn't developed as a Unix-like operating system.

Microsoft didn't start with a completely clean slate, of course. To maintain compatibility with DOS and old Windows software, Windows NT inherited many DOS conventions like drive letters, backslashes for file paths, and forward slashes for command-line switches.

Why it Matters

Have you ever taken a look at the Mac OS X terminal or file system and noticed how similar it was to Linux's, and how different they both were from Windows? Well, this is why — both Mac OSX and Linux are Unix-like operating systems.

Knowing this bit of history helps you understand what a "Unix-like" operating system is, and why so many operating systems seem so similar to each other while Windows seems so different. This explains why the terminal on Mac OS X will feel so familiar to a Linux geek, while the Command Prompt and PowerShell on Windows are so different from other command-line environments.

Unix Features and Advantages

The following are the advantages of Unix Features.

Portability:

The system is written in high-level language making it easier to read, understand, change and, therefore move to other machines. The code can be changed and complied on a new machine. Customers can then choose from a wide variety of hardware vendors without being locked in with a particular vendor.

Machine-independence:

The System hides the machine architecture from the user, making it easier to write applications that can run on micros, mins and mainframes.

Multi-Tasking:

Unix is a powerful multi-tasking operating system; it means when a active task in in process, there can be a simultaneous background process working too. Unix handles these active and background threads efficiently and manages the system resources in a fair-share manner.

Multi-User Operations:

UNIX is a multi-user system designed to support a group of users simultaneously. The system allows for the sharing of processing power and peripheral resources, white at the same time providing excellent security features.

Hierarchical File System:

UNIX uses a hierarchile file structure to store information. This structure has the maximum flexibility in grouping information in a way that reflects its natural state. It allows for easy maintenance and efficient implementation.

UNIX shell:

UNIX has a simple user interface called the shell that has the power to provide the services that the user wants. It protects the user from having to know the intricate hardware details.

Pipes and Filters:

UNIX has facilities called Pipes and Filters which permit the user to create complex programs from simple programs.

Utilities:

UNIX has over 200 utility programs for various functions. New utilities can be built effortlessly by combining existing utilities.

Software Development Tools:

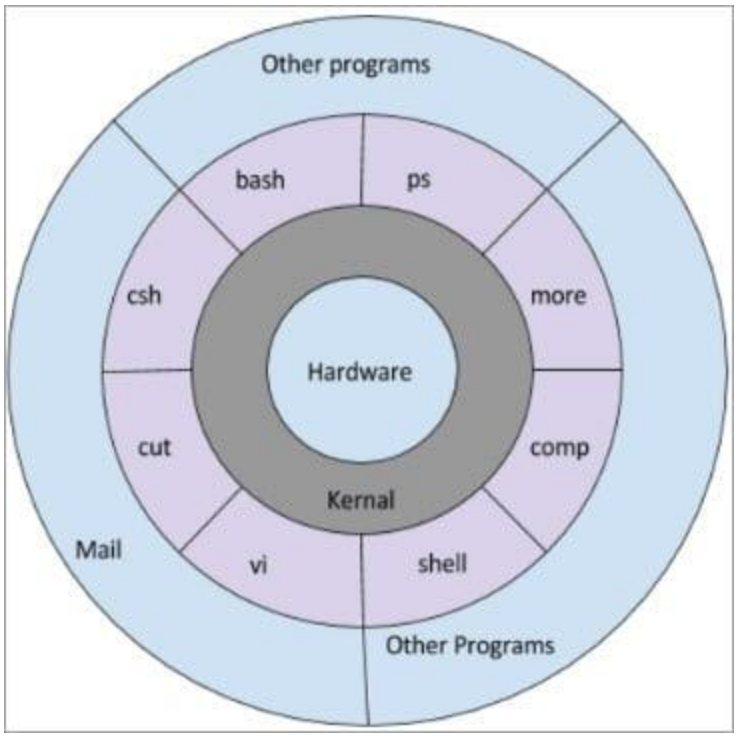
UNIX offers an excellent variety of tools for software development for all phases, from program editing to maintenance of software,

Salient Features of Unix

There are several prominent features of Unix, and few among them are stated below:

- It is a multi-user system where the same resources can be shared by different users.
- It provides multi-tasking, wherein each user can execute many processes at the same time.
- It was the first operating system that was written in a high-level language (C Language). This made it easy to port to other machines with minimum adaptations.
- It provides a hierarchical file structure which allows easier access and maintenance of data.
- Unix has built-in networking functions so that different users can easily exchange information.
- Unix functionality can be extended through user programs built on a standard programming interface.

Unix Architecture



We will understand how user commands are executed in Unix. User commands are often entered on a command line interface that is provided by a 'shell'. The shell is a program that reads the user commands, evaluates it, and then prints the result. For evaluating the command, the shell may execute other commands, or pass them to the 'kernel'.

The kernel is the core of the operating system that directly interacts with the underlying hardware to provide a set of standard services.

UNIT-4(File Management)
Unix Vs Linux
LECTURE NO:-5

Unix Vs Linux: What Is Difference Between UNIX And Linux

nix Vs Linux: Learn what is the Core Difference between UNIX and Linux Architecture, Kernel And Commands

Linux is nothing but a UNIX clone which is written Linus Torvalds from scratch with the help of some hackers across the globe.

Unix and Unix-like operating systems are a family of computer operating systems that derive from the original Unix System from Bell Labs which can be traced back to 1965. Linux is the most popular variant and there comes in a number of different distributions.

Unix is a family of multitasking, portable, multi-user computer operating systems, which also have time-sharing configurations.

Unix systems use a centralized OS kernel which is responsible for managing the entire system.

The programming interface, file abstraction, built-in networking and persistent background processing called daemons are the other features and capabilities that are supported by a Unix OS.

What You Will Learn: [show]

What is UNIX?

Unix is considered as the mother of most of the operating systems.

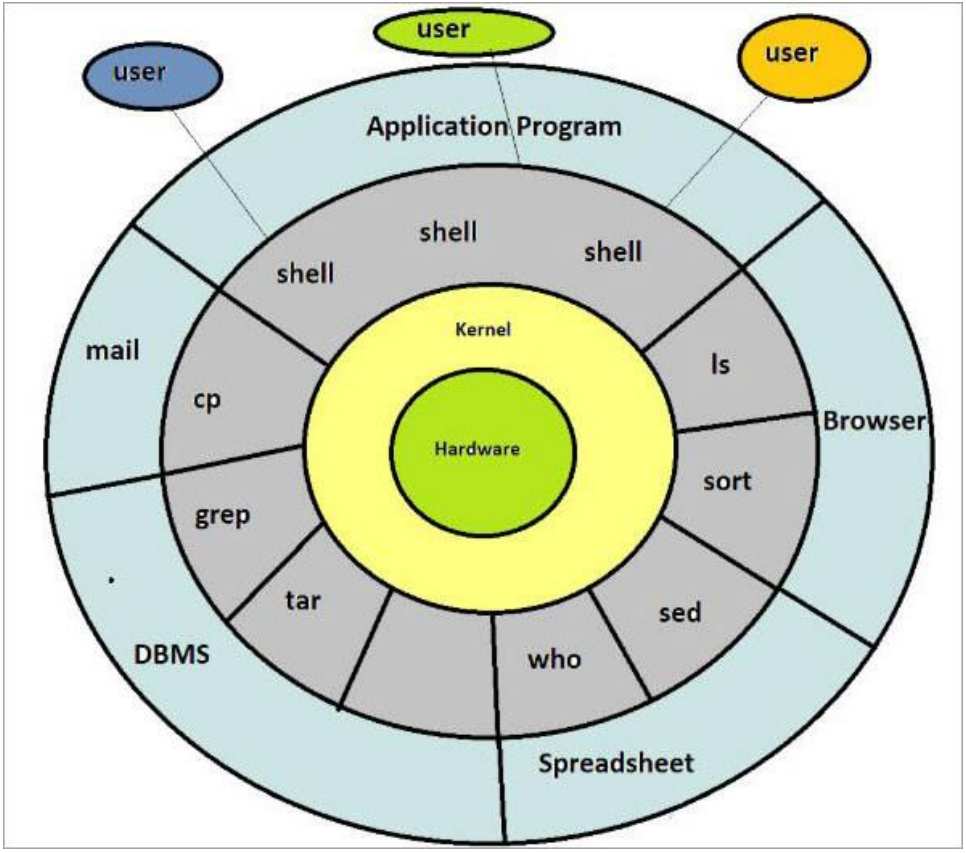
The design of Unix systems is based on "Unix Philosophy" which includes the following characteristics:

- Usage of plain text for data storage.
- Hierarchical file system.
- Handling devices and some specific kinds of inter-process communication (IPC) as files.
- Employing a huge number of software tools.
- Multiple small, simple and modular programs which can be threaded together via a command-line interpreter using pipes, contrasting to use a single monolithic program which comprises of all the same functionality.

It's worth mentioning here the below quote about Unix Philosophy:
"Although that philosophy can't be written down in a single sentence, as its heart is the idea that the power of a system comes more from the relationships among programs than from the programs themselves. Many UNIX programs do quite trivial things in isolation, but, combined with other programs, become general and useful tools." – ***Brian Kernighan & Rob Pike***

Unix Architecture

The below diagram will depict you the Unix architecture.



The master control program of Unix is its Kernel. The kernel has full control over the entire system. It has subsystems that offer services to file system handling, resource handling, memory management, start & stop programs and few other low-level core tasks.

The kernel is the heart of the OS and acts as an interface between the user and hardware. Each kernel subsystem has certain features like concurrency, virtual memory, paging, and a virtual file system.

In the outer layers of the architecture, we have the shell, commands and application programs. Shell is the interface between the user and the kernel. Shell and the user types in the commands, interpret these commands and call the computer programs accordingly.

Example of the Unix operating system is Solaris and HP-UX. The largest distributors of UNIX systems include IBM, HP, and SUN.

Recommended Read => Free Unix Training Tutorials

What is Linux?

By now you would have got a fair idea about Unix. Let's now explore Linux in detail.

People do confuse a lot between the terms Unix and Linux and they generally ask questions like "Is Unix Different from Linux?" / "Are Linux and Unix the same thing?" / "Is Linux like Unix?"/ "Is Linux built on Unix?".

Here is the answer to all such questions. First, let me clear your confusion in a one-liner. Linux and Unix are different but they do have a relationship with each other as Linux is derived from Unix.

Linux is not Unix, but it is a Unix-like operating system. Linux system is derived from Unix and it is a continuation of the basis of Unix design. Linux distributions are the most famous and healthiest example of the direct Unix derivatives. BSD (Berkley Software Distribution) is also an example of a Unix derivative.

At this juncture, it is important for us to make you clear about what is Unix-like?

A Unix-like OS (also called as UN*X or *nix) is the one that works in a way similar to Unix systems, however, it is not necessary that they conform to Single UNIX Specification (SUS) or similar POSIX (Portable Operating System Interface) standard.

SUS is a standard which is required to be met for any OS to qualify for using 'UNIX' trademark. This trademark is granted by 'The Open Group'.

Few Examples of currently registered UNIX systems include macOS, Solaris, and AIX. If we consider the POSIX system, then Linux can be regarded as Unix-like OS.

As per Linux kernel official README file, Linux is a UNIX clone that is developed from scratch by Linus Torvalds and team. It targets for POSIX compliance. The Linux kernel code was completely written from scratch. It is designed in such a way so that it acts like Unix but it does not have the original Unix code in it.

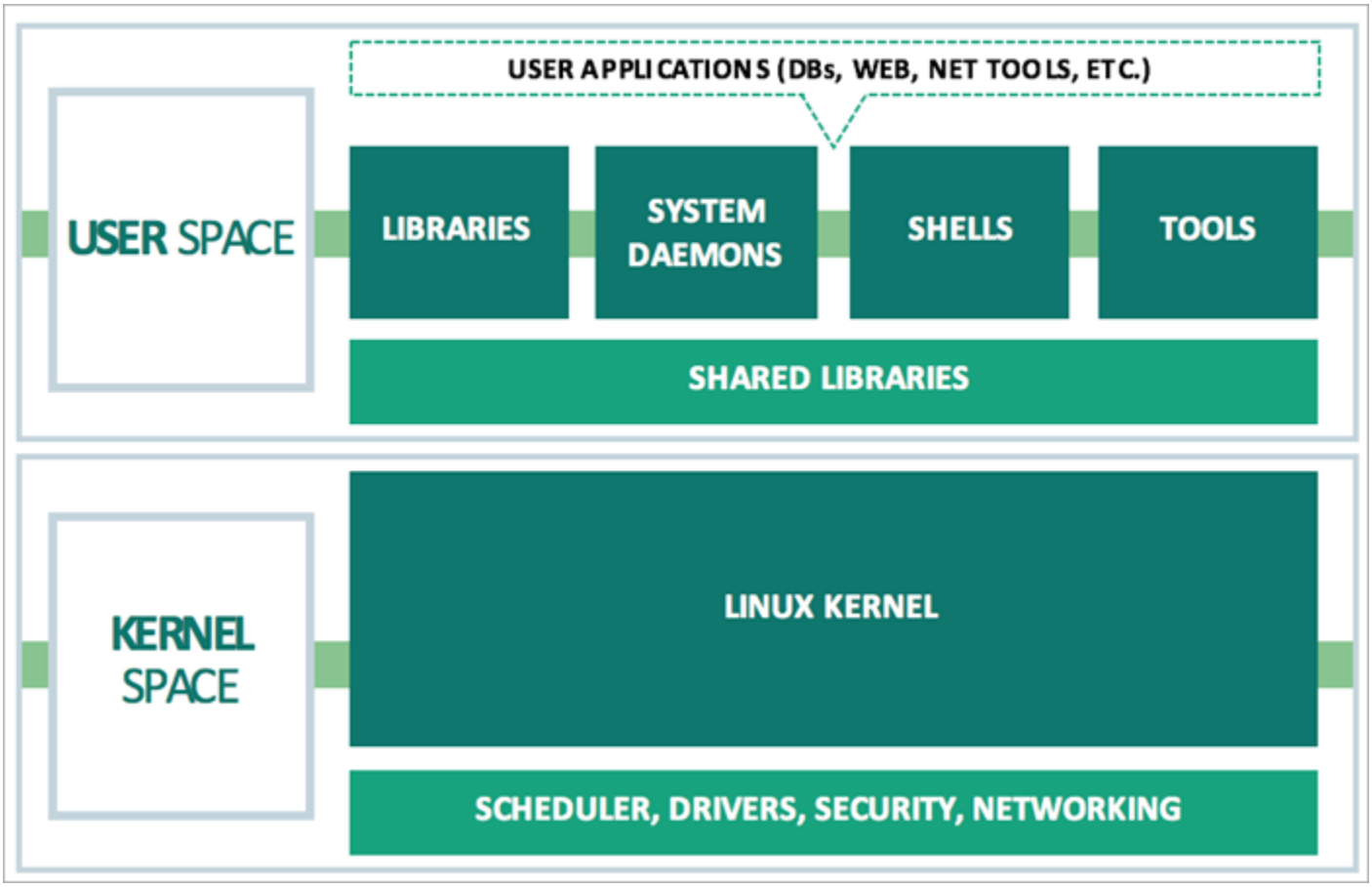
It is also significant to note that Linux is just the kernel and not the complete OS. This Linux kernel is generally packaged in Linux distributions which thereby makes it a complete OS.

Thus, Linux is only the Kernel, while Linux distributions can be treated as the OS. On the other hand, UNIX in itself is a complete OS as everything (all required application tied together) comes from a single vendor. **For Example,** Solaris.
Linux distribution (also called as a distro in short) is an operating system that is created from a collection of software built upon the Linux Kernel and is a package management system.

A standard Linux distribution consists of a Linux kernel, GNU system, GNU utilities, libraries, compiler, additional software, documentation, a window system, window manager and a desktop environment.

Most of the software included in Linux distribution is free and open source. They may include some proprietary software like binary blobs which is essential for a few device drivers.

Linux based OS Architecture



Thus, Linux distributions actually make the Linux kernel completely usable as an operating system by adding different applications to it. There are various flavors of Linux distributions that serve a wide range of user needs.

For Example, we have OpenWrt Linux based OS for embedded devices, Linux Mint for Personal computers and Rocks Cluster Distribution for supercomputers. In total, around 600 Linux distributions do exist.
It will be interesting for you to know that Google's popular Android mobile OS is based on Linux. Every iteration of the Android OS is built on the current Linux kernel.



Difference Between Unix and Linux

Linux	Unix and other Variants
Linux refers to the kernel of the GNU/Linux operating system. More generally, it refers to the family of derived distributions.	Unix refers to the original operating system developed by AT&T. More generally, it refers family of derived operating systems.
Original code developed by Linus and the GNU Foundation	Original code developed by AT & T

Linux	Unix and other Variants
The Linux trademark is owned by Linus Trovalds, and managed by the Linux Mark Institute under the Linux Foundation.	The UNIX trademark is certified by the Open Group. List of certified operating systems.
The Linux Standard Base (LSB), available as ISO/IEC 23360, is a standardization effort by a number of Linux distributors. LSB is mostly an extension of POSIX but has some differences. However, there isn't a strong need for LSB certification as the various distributions use the same kernel in any case.	UNIX certification based on the 'Single Unix Specification' which is an extension of IEEE 1003 (POSIX), also available as ISO/IEC 9945 POSIX specifies programming APIs and shell a utility interfaces. POSIX was developed as a w to allow interoperability between different UNI vendors.
GNU/Linux and derivates like Debian and Fedora	System-V Unix and derivatives like IBM-AIX and HP-UX; Berkeley Unix and derivatives lik FreeBSD and macOS
Open Source under the copyleft General Public License	Berkeley Unix is partially open source under th BSD License. System-V Unix source may be procured under a proprietary commercial licens
Different variants maintained by different communities; with the kernel merging into the branch maintained by Linus	Different variants maintained by different companies; each maintains their own kernel
Designed as a general-purpose scalable platform for a broad set of applications.	Typically designed for a narrow audience with defined set of target platforms and applications
Broadly available as configurable software download and installer.	Typically shipped along with hardware e.g. MacBook
Free community support. Paid support available from a number of service providers.	Paid commercial support. Often leads to vendo lock-in.
Interfaces often evolve	Interfaces usually stable
Frequent updates, with quick bug fixes	Infrequent updates, and fixes may take time
Supports almost all file systems used across operating systems	Most versions support two or perhaps three file systems

Linux	Unix and other Variants
Breadth of system administration tools often with limited focus e.g. Suse YAST	Each version typically has a mature system administration tool e.g. HP SAM
Preferred OS for cloud deployment and data centers primarily for economic reasons	Preferred OS for special purpose server requirements due to application availability, an internet servers for legacy reasons
Scalability achieved using clusters, grids or cloud.	Scalability achieved using clusters or grids
(A cluster is a collection of homogenous computers, a grid is a collection of distributed computers, and a cloud service is a collection of virtualized clusters.)	
Most of the command line and graphical utilities are similar to Unix	Most of the command line and graphical utilitie are similar to Linux

We hope you must have understood the core differences between Unix and Linux from this article.

Let us now see some more important differences between Linux and Unix in the below tabular format:

Features	Linux	Unix
Developer	Inspired by MINIX (a Unix-like OS), Linux was originally developed by Finnish-American software engineer Linus Torvalds. Since it is an open source, we have community developers for Linux.	Originally derived from AT&T Uni it was developed at Bell Labs by Kenneth Lane Thompson, Dennis Ritchie, and 3 others.
Written in	C and other programming languages.	C and assembly language.
OS family	Unix-like	Unix
Working state	Current	Current
Source Model	Open source	Mixed. Traditionally closed source,

Features	Linux	Unix
		however, few Unix projects are ope source which include illumos OS a BSD (Berkley Software Distributio OS.
Available in	Multilingual	English
Initial release	Linux is newer when compared to Unix. It was derived from Unix and was released in September 1991.	Unix is older. Was released in October 1973 for outside parties. Before that, it was used internally i Bell Labs since its inception in 197
Kernel Type	Monolithic kernel	Kernel Type varies. It can be monolithic, microkernel and hybrid
License	GNUv2(GPL General Public License) and others.	Licensing varies. Few versions are proprietary while others are free/O
Official Website	https://www.kernel.org/	http://opengroup.org/unix
Default user interface	Unix shell	CLI (Command Line Interface) and Graphical (X Windows system)
Text Mode Interface	By default, the shell is BASH (Bourne Again Shell). Moreover, is compatible with many command interpreters.	Originally the Bourne shell. It is als compatible with many command interpreters.
Cost	Can be obtained and used freely. There are priced versions of Linux as well. But, generally, Linux is cheaper than Windows.	Proprietary operating systems have different cost structures set accordingly by the vendors selling
Examples	Debian, Ubuntu, Fedora, Red Hat, Android, etc.	IBM AIX, Solaris, HP-UX, Darwin macOS X, etc.
Architecture	Was originally created for Intel's x86 hardware, ports available for a lot of CPU types.	Compatible with PA and Itanium machines. Solaris is also available x86/x64. OSX is PowerPC.
Threat	As Linux is mainly driven by open source	Due to the proprietary nature of Un

Features	Linux	Unix
detection and solution	community, many developers across different parts of the world are working on the code. Hence threat detection and solution is quite fast in case of Linux.	users need to wait for proper bug fixing patches.
Security	Both Linux and Unix based OS is generally regarded as very well protected against malware. This is attributable to lack of root access, quick updates and comparatively low market share (as compared to windows). As of 2018, there has been none widespread Linux virus.	Unix is also considered to be very safe. It is even harder to infect as th source is also not available. There i no actively spreading virus for Unix nowadays.
Price	Linux is free. However, corporate support is available at a price.	Unix is not free. However, some Un versions are free for development u (Solaris). In a collaborative environment, Unix costs $1,407 per user and Linux costs $256 per user. Hence, UNIX is extremely expensi

Examples

So far, in this article, we have seen the generalized core differences between Linux and Unix. These differences can be more specific if we compare the exact versions of the two. Let us see this through some examples.

Solaris vs Linux

Solaris, which is now called Oracle Solaris is a Unix family OS. Let's compare Linux with Solaris.

Linux supports more system architectures than Solaris does. Hence, Linux is more portable.

While talking about stability and hardware integration, Solaris seems to better here. Linux also has a faster rate of development when compared to Solaris.

There are few other technical differences between the two, but here we are limiting our comparison only to performance.

MacOS vs Linux

MacOS is a certified Unix OS. It has its own kernel named XNU. It is used in Apple's computers which are considered as the most reliable PCs.

MacOS is relatively easy to set up. On the other side, Linux is cheaper and has a lot of open source software available as against Apple's proprietary solutions. Also, Linux is more flexible as it can be executed on almost any hardware whereas MacOS can run only on Apple hardware. **For Example**, iPhones.
MacOS uses HFS+ as a default file system whereas Linux uses ext4.

Conclusion
Unix is very old and is said to be the mother of all operating systems. Linux kernel is also derived from Unix. The major difference between Unix and Linux based operating systems is not in the presentation part, but on how they work internally, i.e. mainly at the kernel part.

The difference between the two will also depend upon which exact versions of Linux and Unix you are comparing.

It's also essential to state that Linux (and many other Unix-like OS) are free to obtain and modify, whereas Unix operating systems are not. Cost is always a major concern while deciding what technology to use, and Linux has an edge in this regard.

Linux is more flexible and free when compared to true Unix systems and that is why Linux has gained more popularity. While discussing the commands in Unix and Linux, they are not the same but are very much similar. In fact, the commands in each distribution of the same family OS also varies.

Solaris, HP, Intel, etc. employ Unix internet servers, workstations, and personal computer. While, Linux is widely employed for computer software & hardware, gaming, tablet, mainframes, etc.

There are studies which say that Linux is growing fast than any other OS in the past few years. Hence, in the future, Linux may tend to leave UNIX installations far behind.

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UNIT-4(File Management)
Mobile Operating System
LECTURE NO:-6

What is a Mobile Operating System (Mobile OS)?

Much like the Linux or Windows operating system controls your desktop or laptop computer, a mobile operating system is the software platform on top of which other programs can run on mobile devices. The operating system is responsible for determining the functions and features available on your device, such as thumb wheel, keyboards, WAP, synchronization with applications, email, text messaging and more. The mobile OS will also determine which third-party applications (mobile apps) can be used on your device.

Types of Mobile Operating Systems

When you purchase a mobile device the manufacturer will have chosen the operating system for that specific device. Often, you will want to learn about the mobile operating system before you purchase a device to ensure compatibility and support for the mobile applications you want to use.

9 Popular Mobile Operating Systems

1. Android OS (Google Inc.)

The Android mobile operating system is Google's open and free software stack that includes an operating system, middleware and also key applications for use on mobile devices, including smartphones. Updates for the open source Android mobile operating system have been developed under "dessert-inspired" version names (Cupcake, Donut, Eclair, Gingerbread, Honeycomb, Ice Cream Sandwich) with each new version arriving in alphabetical order with new enhancements and improvements.

2. Bada (Samsung Electronics)

Bada is a proprietary Samsung mobile OS that was first launched in 2010. The Samsung Wave was the first smartphone to use this mobile OS. Bada provides mobile features such as multipoint-touch, 3D graphics and of course, application downloads and installation.

Did You Know…? *In the computer industry, proprietary is the opposite of open. A proprietary design or technique is one that is owned by a company. It also implies that the company has not divulged specifications that would allow other companies to duplicate the product.*

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3. BlackBerry OS (Research In Motion)

The BlackBerry OS is a proprietary mobile operating system developed by Research In Motion for use on the company's popular BlackBerry handheld devices. The BlackBerry platform is popular with corporate users as it offers synchronization with Microsoft Exchange, Lotus Domino, Novell GroupWise email and other business software, when used with the BlackBerry Enterprise Server.

4. iPhone OS / iOS (Apple)

Apple's iPhone OS was originally developed for use on its iPhone devices. Now, the mobile operating system is referred to as iOS and is supported on a number of Apple devices including the iPhone, iPad, iPad 2 and iPod Touch. The iOS mobile operating system is available only on Apple's own manufactured devices as the company does not license the OS for third-party hardware. Apple iOS is derived from Apple's Mac OS X operating system.

5. MeeGo OS (Nokia and Intel)

A joint open source mobile operating system which is the result of merging two products based on open source technologies: Maemo (Nokia) and Moblin (Intel). MeeGo is a mobile OS designed to work on a number of devices including smartphones, netbooks, tablets, in-vehicle information systems and various devices using Intel Atom and ARMv7 architectures.

6. Palm OS (Garnet OS)

The Palm OS is a proprietary mobile operating system (PDA operating system) that was originally released in 1996 on the Pilot 1000 handheld. Newer versions of the Palm OS have added support for expansion ports, new processors, external memory cards, improved security and support for ARM processors and smartphones. Palm OS 5 was extended to provide support for a broad range of screen resolutions, wireless connections and enhanced multimedia capabilities and is called Garnet OS.

7. Symbian OS (Nokia)

Symbian is a mobile operating system (OS) targeted at mobile phones that offers a high-level of integration with communication and personal information management (PIM) functionality. Symbian OS combines middleware with wireless communications through an integrated mailbox and the integration of Java and PIM functionality (agenda and contacts). Nokia has made the Symbian platform available under an alternative, open and direct model, to work with some OEMs and the small community of platform

development collaborators. Nokia does not maintain Symbian as an open source development project.

8. webOS (Palm/HP)

WebOS is a mobile operating system that runs on the Linux kernel. WebOS was initially developed by Palm as the successor to its Palm OS mobile operating system. It is a proprietary Mobile OS which was eventually acquired by HP and now referred to as webOS (lower-case w) in HP literature. HP uses webOS in a number of devices including several smartphones and HP TouchPads. HP has pushed its webOS into the enterprise mobile market by focusing on improving security features and management with the release of webOS 3.x. HP has also announced plans for a version of webOS to run within the Microsoft Windows operating system and to be installed on all HP desktop and notebook computers in 2012.

9. Windows Mobile (Windows Phone)

Windows Mobile is Microsoft's mobile operating system used in smartphones and mobile devices – with or without touchscreens. The Mobile OS is based on the Windows CE 5.2 kernel. In 2010 Microsoft announced a new smartphone platform called Windows Phone 7.

Did You Know...? According to data from the International Data Corporation (IDC) worldwide smartphone shipments declined 0.5% in 2017, the first year-over-year decline the market has experienced since the introduction of what we now know as smartphones. Smartphone companies shipped a total of 1.46 billion devices in 2017 with nearly all of that volume running either the Android or iOS platforms.